EE3032 Introduction to VLSI Design – Final

Fall 2007

1. (26%) Answer the following problems.

(a) List major steps of the full-custom design flow. (8%)

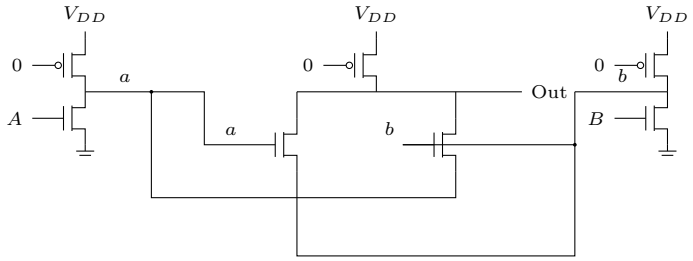
(b) List two basic rules for drawing the layout of a standard cell. (4%)

(c) List two major problems of dynamic logic. (4%)

(d) List two limitations of CMOS domino logic. (4%)

(e) What are core-limited design and pad-limited design? (6%)

2. (10%) What logic function does the circuit of Figure 1 realize? Explain your answer in detail.



All three PMOS gates are at VSS (always on).
Wire crossings without a dot are not connected.

Figure 1: A logic circuit. Added labels a, b identify the intermediate nodes.

3. (14%) Figure 2 shows a layout view of a CMOS circuit. Draw its transistor schematic diagram.

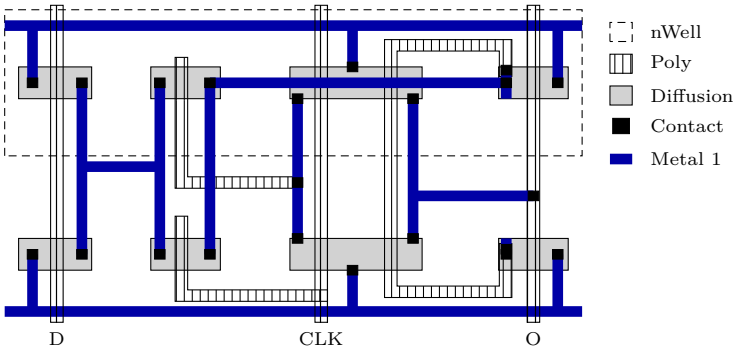


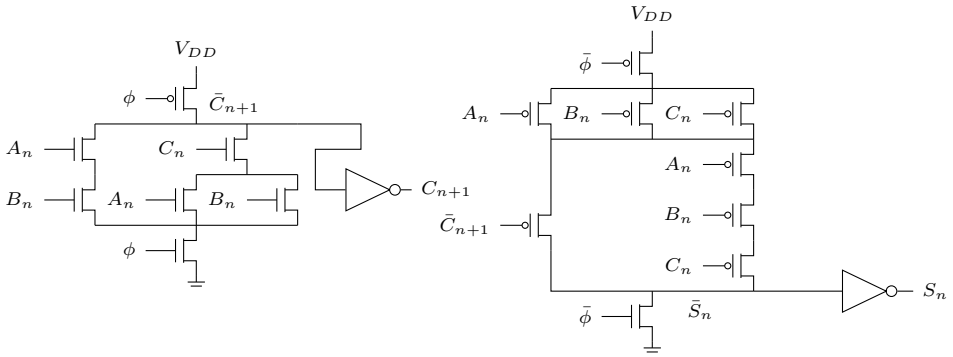
Figure 2: Symbolic layout of a CMOS circuit. Geometry normalized; contacts and split poly preserved.

4. (20%) The circuit in Figure 3 implements a high-speed one-bit full adder. Complete the following problems.

(a) What type of logic was used to implement this circuit? (5%)

(b) Describe the operation of the circuit. (5%)

(c) The drain and gate capacitances of PMOS and NMOS are 0.2 fF and 1 fF, respectively. The leakage current of output nodes $\bar{C}_{n+1}$ and $\bar{S}_n$ is 1 nA. Also $V_{DD} = 1.8$ V and the inverter switching threshold is $V_{DD}/2$. Determine the lowest operating speed of the circuit. (10%)



The two receiving triangles are CMOS inverters.
Labels preserve the source's $\phi/\bar{\phi}$ connections.

Figure 3: An adder circuit.

5. (15%) Sketch $f = \overline{A(B+C)+DE}$ using each of: (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%).

6. (10%) Explain the operation principle of Figure 4.

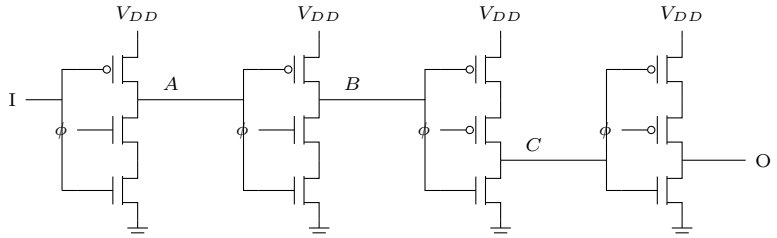


Figure 4: A CMOS circuit.

7. (10%) Figure 5 shows a pipelined sequential circuit. Combinational propagation delay is 15 ns. Clock-to-Q delay, setup time, and hold time of R1/R2 are 2 ns, 3 ns, 1 ns, respectively. Determine the CLK period for the highest operation speed.

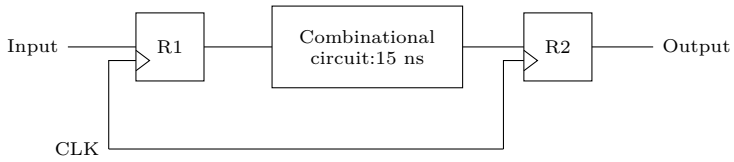


Figure 5: A pipelined circuit.

Archive note: This editable reconstruction covers the dated Fall 2007 paper. The printed percentages total 105%; none is rescaled. Source annotations are distinguished from original printed circuit connections. All restored diagrams are editable vectors.